

Gaussian Channels and the True Covariance

How the simulation constructs T , $X1$, and $X2$

This note explains the Gaussian channel used by `make_both_unique_true_cov_from_config`, how its population covariance is obtained, and why a connection probability of zero creates exactly zero mutual information between every source and the target.

The generative channel

Let the target be a standard Gaussian vector. Each source observes a linear transformation of the target plus independent unit Gaussian noise:

$$T \sim N(0, I_t)$$

$$X1 = H1 T + e1, \quad e1 \sim N(0, I_1)$$

$$X2 = H2 T + e2, \quad e2 \sim N(0, I_2).$$

T , $e1$, and $e2$ are mutually independent. $H1$ and $H2$ are the two channel matrices. Entry (i,j) is one when target coordinate j connects to source coordinate i , and zero otherwise. Every entry is sampled independently:

$$H1(i, j), H2(i, j) \sim \text{Bernoulli}(\pi).$$

The YAML variable `both_unique_connection_probability` supplies π . The value 0.1 gives the sparse-channel construction used for the both-unique example. The value 0.0 makes both channel matrices exactly zero.

Deriving the covariance blocks

All variables have zero mean. Using the channel equations, independence, and $\text{Cov}(T)=I_t$:

$$\text{Cov}(X1) = H1 H1^T + I_1$$

$$\text{Cov}(X2) = H2 H2^T + I_2$$

$$\text{Cov}(X1, X2) = H1 H2^T$$

$$\text{Cov}(X1, T) = H1$$

$$\text{Cov}(X2, T) = H2.$$

For the project ordering $[X1, X2, T]$, these blocks form the true population covariance:

$$\text{Sigma} = \begin{bmatrix} H1 H1^T + I_1 & H1 H2^T & H1 \\ H2 H1^T & H2 H2^T + I_2 & H2 \\ H1^T & H2^T & I_t \end{bmatrix}.$$

For example, $\text{Cov}(X1, T)$ follows from

$$\begin{aligned} \text{Cov}(H1 T + e1, T) \\ &= H1 \text{Cov}(T, T) + \text{Cov}(e1, T) \\ &= H1 I_t + 0 \\ &= H1. \end{aligned}$$

The source-source covariance $H1 H2^T$ does not mean that one source directly causes the other. It appears because both sources observe the same target. The added identity matrices are the covariances of the independent source noises.

Why $\pi = 0$ gives exact zero information

When $\pi=0$, a Bernoulli draw is always zero. Therefore $H_1=0$ and $H_2=0$ deterministically. The covariance reduces to

$$\Sigma = \text{diag}(I_1, I_2, I_t).$$

All source-target cross-covariance blocks are exactly zero. For jointly Gaussian variables, zero cross-covariance implies independence. Thus

$$I(X_1; T) = 0,$$

$$I(X_2; T) = 0,$$

$$I([X_1, X_2]; T) = 0.$$

The joint-source result is important: stacking the two source-target blocks still gives a zero block, so combining the sources cannot reveal target information. All true PID atoms are consequently zero.

Finite samples generally have small nonzero empirical cross-covariances. Their PID estimates can therefore be nonzero even though the population truth is zero. Those errors are the finite-sample bias and variance that the Venkatesh and Lorenz corrections are intended to reduce.

Configuration

Use the following settings in either simulation YAML:

```
covariance:
  use_both_unique_covariance: true
  both_unique_connection_probability: 0.0
```

Change 0.0 to 0.1 to restore the nonzero sparse-channel example. While `use_both_unique_covariance` is true, the direct-covariance parameters `p_scale`, `q_scale`, and `r_scale` are not used to construct the covariance.

Population truth versus sampled covariance

The function returns the population covariance above. The simulation first uses that matrix to calculate the theoretical PID truth and then draws finite Gaussian samples from the same matrix. The theoretical covariance does not change across sample trials; only the estimated sample covariance changes.

References

- Venkatesh et al., "Gaussian Partial Information Decomposition: Bias Correction and Application to High-dimensional Data," NeurIPS 2023, Appendix C.1.
- Lorenz et al., "Sampling bias corrections for discrete and Gaussian partial information decompositions," Patterns, 2026.